

# Template Week 5 – Operating Systems

Student number: 530010

## Assignment 5.1: Unix-like

- a) Find out what the difference is between UNIX and unix-like operating systems?  
All UNIX systems are Unix-like, but not all Unix-like systems are UNIX.
- b) Study the image above named UNIX timeline. Find out who Ken Thompson, Dennis Ritchie, Bill Joy, Richard Stallman, and Linus Torvalds are and what they have contributed to the development of UNIX or unix-like systems and to IT in general. **TIP!** English-language sources often contain more detailed information about these individuals.

Ken Thompson

- Co-creator of UNIX at Bell Labs (late 1960s).
- Designed the original UNIX kernel.
- Created the B programming language, precursor to C.
- Major influence on operating system design.

Dennis Ritchie

- Co-creator of UNIX with Ken Thompson.
- Creator of the C programming language.
- Rewrote UNIX in C, making it portable across hardware.
- One of the most influential figures in computer science.

Bill Joy

- Key developer of BSD UNIX (Berkeley Software Distribution).
- Creator of important UNIX tools: vi editor, csh (C shell).
- Co-founder of Sun Microsystems, creator of Solaris.
- Influential in networking (TCP/IP integration in UNIX).

Richard Stallman

- Founder of the GNU Project (1983).
- Creator of the Free Software Movement.
- Wrote major GNU tools: GCC, Emacs, GDB.
- Author of the GNU General Public License (GPL).
- Advocates software freedom as an ethical issue.

Linus Torvalds

- Creator of the Linux kernel (1991).
- Enabled a complete free Unix-like OS when combined with GNU tools.
- Creator of Git, the distributed version control system.
- Central figure in open-source development.

- c) What is the philosophy of the GNU movement?  
The GNU philosophy is based on software freedom, defined by four essential freedoms:
1. Freedom to run the program for any purpose
  2. Freedom to study how the program works and modify it
  3. Freedom to redistribute copies
  4. Freedom to distribute modified versions

Key principles:

- Software should be free as in freedom, not price
- Source code must be available
- Proprietary software is considered harmful to user freedom
- Copyleft (GPL) ensures freedom remains protected

d) Does Ubuntu as a Linux operating system conform to the philosophy of the GNU movement?

Please explain your answer.

Partially, but not completely.

Yes, because:

- Ubuntu is based on GNU/Linux
- Uses GNU tools (bash, gcc, coreutils)
- Core system is open source
- Encourages free software

No (from a strict GNU viewpoint), because:

- Ubuntu includes proprietary software (drivers, codecs)
- Offers non-free software repositories
- Prioritizes usability over strict software freedom

Conclusion:

Ubuntu is GNU/Linux-based but does not fully conform to the GNU philosophy as defined by Richard Stallman. GNU prefers distributions like Trisquel or PureOS.

e) Find out what is the Windows Subsystem for Linux?

WSL is a compatibility layer in Windows 10 and 11 that allows users to:

- Run Linux distributions directly on Windows
- Use Linux command-line tools and applications
- Access files between Windows and Linux environments

Two versions:

- WSL 1: Translation layer (no real kernel)
- WSL 2: Runs a real Linux kernel in a lightweight virtual machine

Purpose:

- Improve developer productivity
- Allow Linux tools without dual-boot or full VM

f) Find out, which operating system family belongs to Android, iOS and ChromeOS?

| Operating System | OS Family                                  |
|------------------|--------------------------------------------|
| Android          | Linux-based (uses Linux kernel)            |
| iOS              | Unix-based (Darwin / BSD UNIX)             |
| ChromeOS         | Linux-based (Gentoo-derived, Linux kernel) |

Additional notes:

- Android is Linux, but not GNU/Linux
- iOS is a certified UNIX system
- ChromeOS blends Linux with web-centric design

## Assignment 5.2: Supercomputers and gameconsoles

- a) Research on this site what supercomputers are used for and write a short summary of it:

<https://www.computerhistory.org/timeline/search/?q=Supercomputer>

Supercomputers are extremely powerful computers designed to perform massive amounts of calculations much faster than ordinary computers. They are used in areas where huge datasets and complex computations are required, such as:

- Scientific research — climate modeling, astrophysics, fluid dynamics
- Weather forecasting
- Simulating nuclear weapons and stockpile stewardship
- Genomics and drug discovery
- Artificial intelligence and machine learning research
- Large-scale data analysis

Supercomputers differ from normal computers by having many processors working in parallel to achieve extremely high performance. Historically they have been used for government, scientific, and academic research that demands the highest computing power available.

- b) IBM is a company that has already built a number of supercomputers. One of them is IBM's Roadrunner. The CPU developed for this supercomputer was further developed at a later stage as the CPU for the PlayStation 3 console. Find out what a **PlayStation 3 cluster** is and what it was used for?

A PlayStation 3 cluster is a parallel computing system made by linking many PS3 consoles together so they can work on computational tasks as a group. These clusters exploit the PS3's Cell Broadband Engine CPU, which has multiple processing units suitable for high-performance number crunching.

Uses of PS3 clusters historically:

- Research computing at universities and labs
- Running distributed scientific applications using Linux
- The U.S. Air Force built the "Condor Cluster" from 1,760 PS3s to perform tasks like image processing, pattern recognition, radar simulation and AI research — at a fraction of the cost and energy of a traditional supercomputer.

So, a PS3 cluster was essentially a low-cost supercomputing platform that used many gaming consoles to achieve high parallel performance.

- c) You can build a supercomputer by putting a few computers together in a cluster. Here's what Oracle did with a collection of Raspberry Pi's, for example:

<https://blogs.oracle.com/developers/post/building-the-worlds-largest-raspberry-pi-cluster>

What specific operating system is running on this cluster?

The Oracle Raspberry Pi cluster runs Oracle Linux for ARM, which is Oracle's version of the Linux operating system adapted to work on the ARM architecture used by Raspberry Pi boards. This cluster also uses network booting, so all nodes share a common OS image and configuration.

- d) Does Oracle's Raspberry Pi supercomputer appear in the list of the 500 fastest supercomputers in the world? Make a logical decision for this, without going through the entire list.

<https://www.top500.org/lists/top500/list/2023/06/>

No, it almost certainly does *not* appear in the Top500 list of the fastest supercomputers.

Reasoning:

- The Top500 ranks systems by actual high-performance computing (HPC) performance in FLOPS (floating-point operations per second), using Linpack benchmarks.
- A Raspberry Pi cluster — even with hundreds or thousands of nodes — uses low-power ARM CPUs and cannot compete with modern petaflop-class systems (exascale machines like El Capitan).
- Even very large clusters of Raspberry Pis are orders of magnitude slower than the performance required to be in the top500.

So, it is logical that Oracle's Raspberry Pi supercomputer would not appear on the Top500 list. (Because performance is far too low, despite being large in node count.)

- e) What CPU architecture is used for the PlayStation 5 and Xbox Series X?

CPU Architecture

- PlayStation 5: Uses a custom AMD Zen 2-based CPU (64-bit x86-64 architecture).
- Xbox Series X: Also uses a custom AMD Zen 2-based CPU (64-bit x86-64 architecture).

Both consoles use x86-64 architecture (similar to most desktop PCs).

What operating systems run on these consoles?

Operating Systems

- PlayStation 5: Runs a proprietary Sony OS built on a custom real-time OS layer optimized for gaming.
- Xbox Series X: Runs a custom Windows-based OS derived from the Windows kernel and Xbox system software.

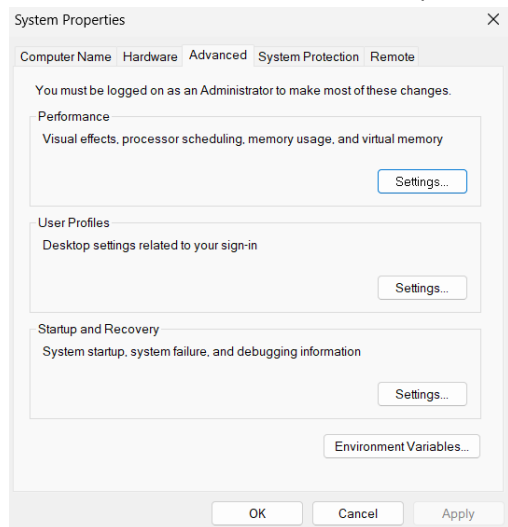
What conclusion can you draw from the answer to the previous question?

Both modern consoles (PS5 & Xbox Series X) use x86-64 CPU architectures and run specialized operating systems tailored for gaming. This shows a trend: modern high-performance consumer devices increasingly share architecture and OS design with general-purpose computing, blurring the line between dedicated systems and general computing platforms.

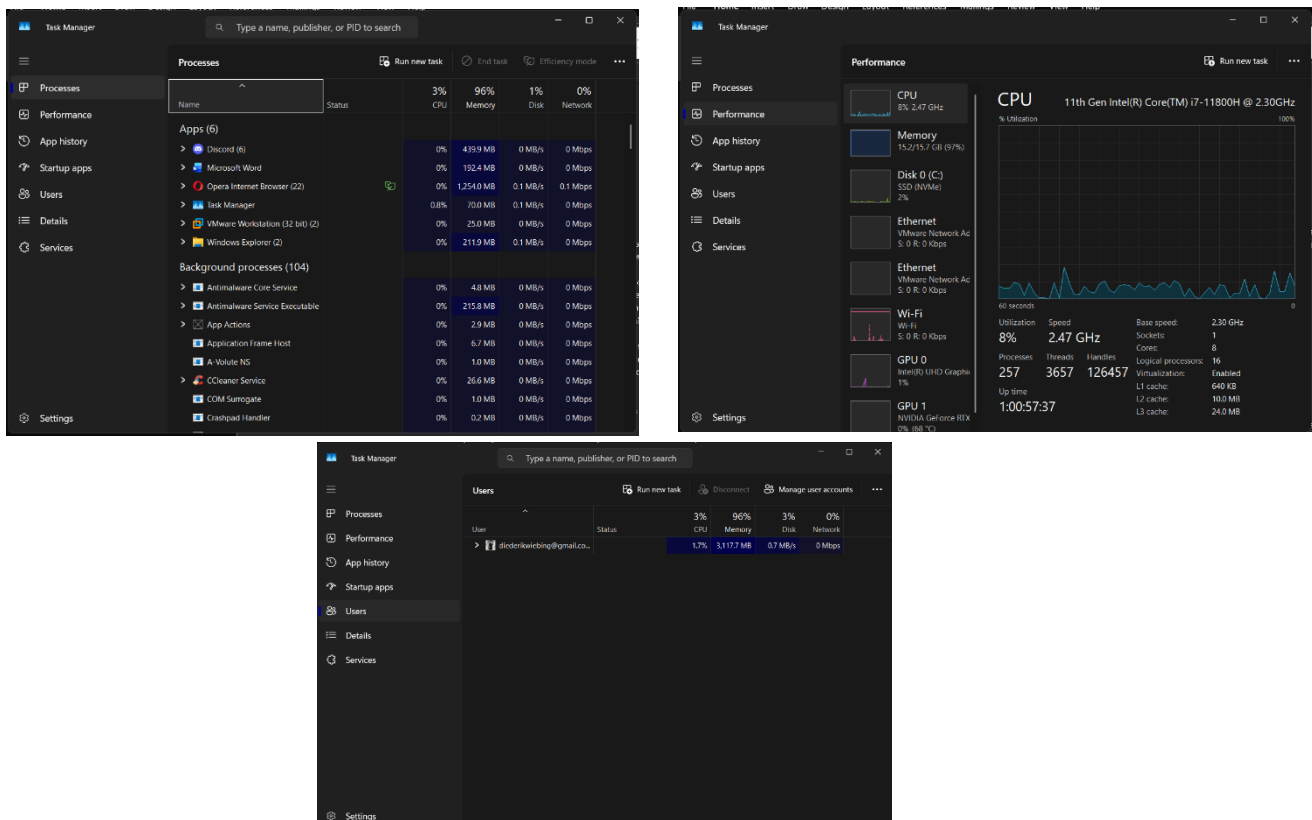
## Assignment 5.3: Working with Windows

### Take relevant screenshots of the assignments below

- Practice for about 10 minutes with the  keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.
- The file explorer can be opened with  + E, Which key combination could you also use?  
Windows + Cijfer
- Open the system properties with a  key combination, take a screenshot of the open screen.  
Paste this screenshot into this template.



- Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.



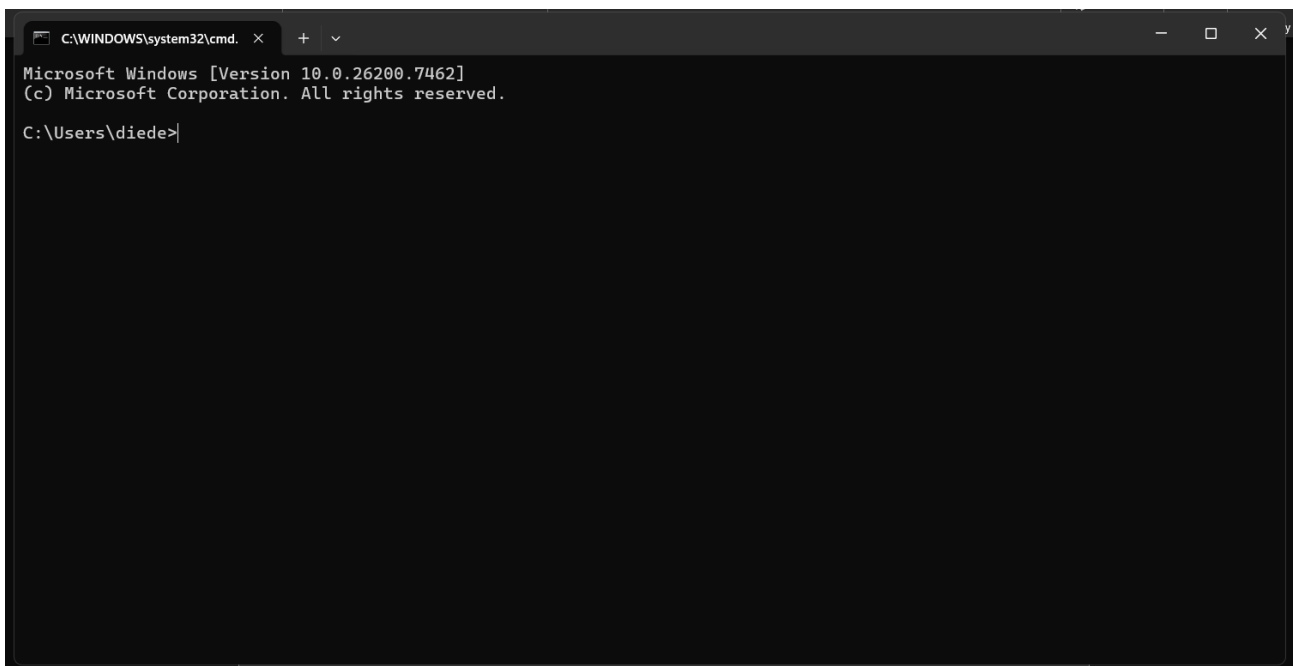
- e) If you're giving a PowerPoint presentation and you connect your laptop to a projector, Windows can use the projector as a second screen. For example, you may have Outlook open on your first screen that you don't show over the projector, while the PowerPoint presentation is displayed on the projector, or the second screen. Which key combination should you use for this?

Windows + P

- f) If you leave the classroom for a while and you leave your laptop behind, it is wise to lock the screen. Your Apps will continue to run in the background. So, for example, if you're waiting for a download that takes a while, lock the screen and get a cup of coffee. Which key combination do you use for this?

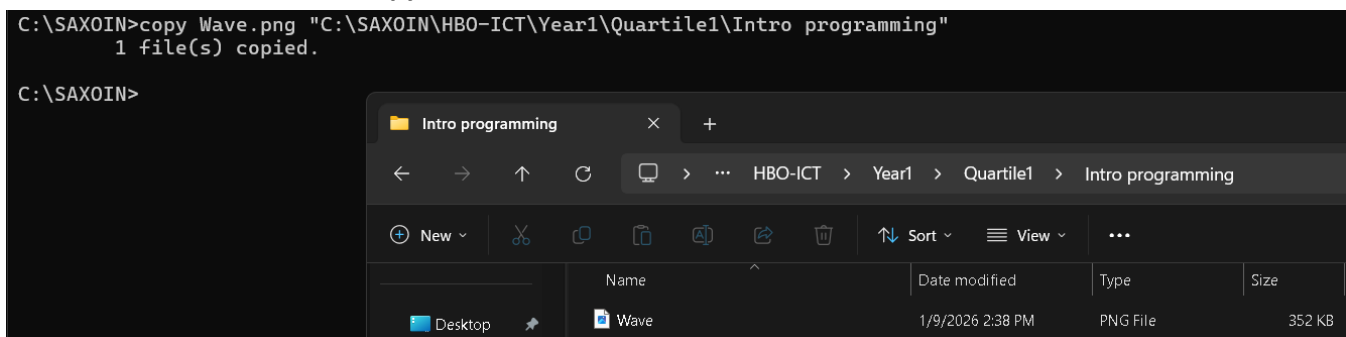
Windows + L

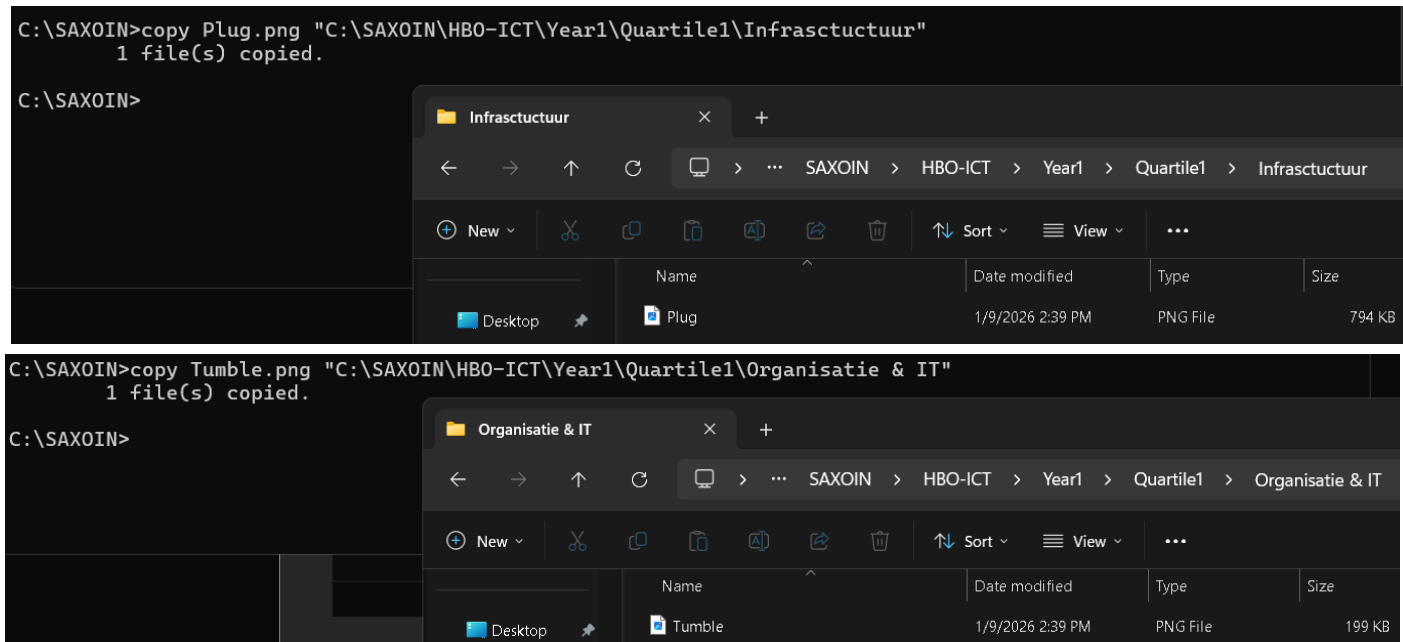
- g) Open the Run screen with a key combination. On this screen, type CMD and press <enter>. Take a screenshot of this result and paste it into this template.



## Working in the File Explorer

Relevant screenshots **copy** command:

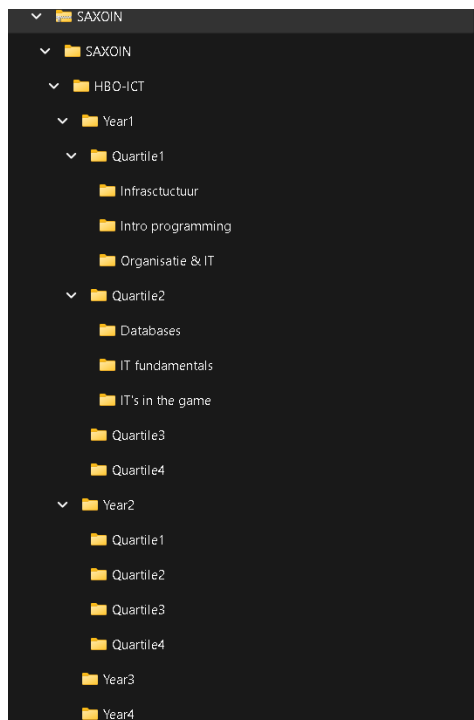




Relevant screenshots **tree** command:

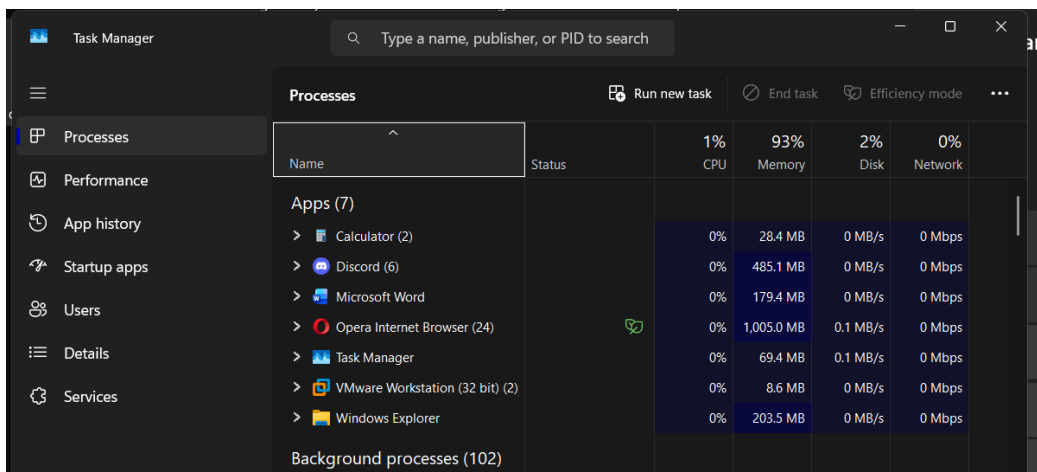
```
C:\SAXOIN>tree
Folder PATH listing for volume Windows
Volume serial number is 3A7F-B760
C:..
├── HBO-ICT
│   ├── Year1
│   │   ├── Quartile1
│   │   │   ├── Infrasctuctuur
│   │   │   ├── Intro programming
│   │   │   └── Organisatie & IT
│   │   ├── Quartile2
│   │   │   ├── Databases
│   │   │   ├── IT fundamentals
│   │   │   └── IT's in the game
│   │   ├── Quartile3
│   │   └── Quartile4
│   ├── Year2
│   │   ├── Quartile1
│   │   ├── Quartile2
│   │   ├── Quartile3
│   │   └── Quartile4
│   ├── Year3
│   └── Year4
C:\SAXOIN>echo %diederik%
%diederik%
```

Relevant screenshots in the file explorer of the folder c:\Saxion + created zip file.



## Terminating Processes

Relevant Screenshots Task Manager Window:





## Install Software

Relevant screenshots that the following software is installed with winget:

- WinSCP
- Notepad++
- 7zip

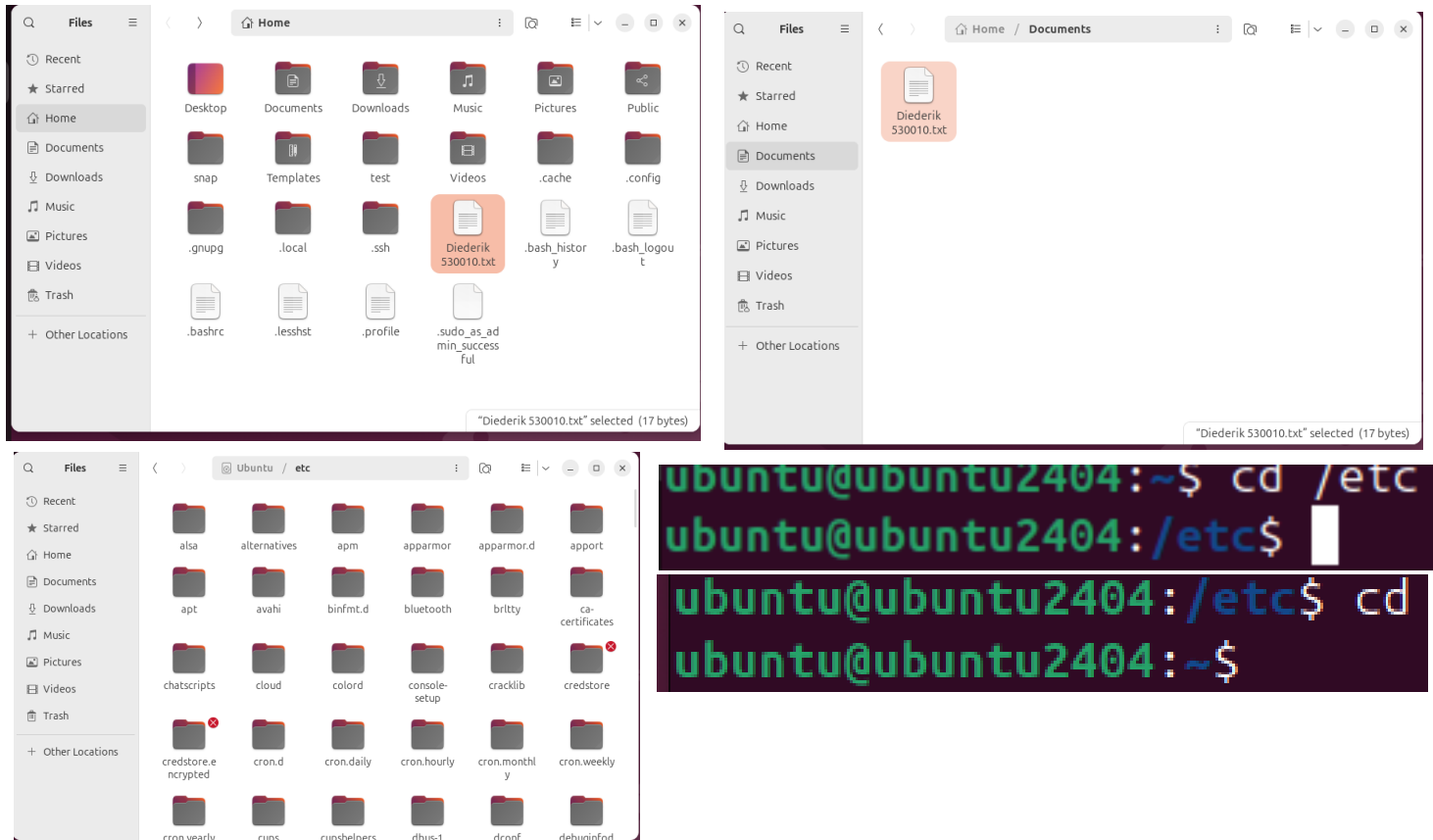
```
C:\Windows\System32>winget install 7Zip
Found an existing package already installed. Trying to upgrade the installed package...
No available upgrade found.
No newer package versions are available from the configured sources.
```

```
C:\Windows\System32>winget install notepad++  
Found an existing package already installed. Trying to upgrade the installed package...  
Found Notepad++ [Notepad++.Notepad++] Version 8.9  
This application is licensed to you by its owner.  
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.  
Downloading https://github.com/notepad-plus-plus/notepad-plus-plus/releases/download/v8.9/npp.v8.9.Installer.x64.exe  
██████████ 6.54 MB / 6.54 MB  
Successfully verified installer hash  
Starting package install...  
Successfully installed
```

```
C:\Windows\System32>winget install winscp
Found WinSCP [WinSCP.WinSCP] Version 6.5.5
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://sourceforge.net/projects/winscp/files/WinSCP/6.5.5/WinSCP-6.5.5-Setup.exe/download
██████████ 11.6 MB / 11.6 MB
Successfully verified installer hash
Starting package install...
Successfully installed
```

## Assignment 5.4: Working with Linux

### Relevant screenshots + motivation



A significant difference between Linux and Windows file structures is:

- Linux uses a unified hierarchical directory structure starting from / (root), whereas Windows uses separate drive letters like C:\, D:\, etc.  
In Linux, everything (system files, user files, devices) is organized under the single root /, while in Windows, each drive has its own independent structure.

As for /etc in Linux:

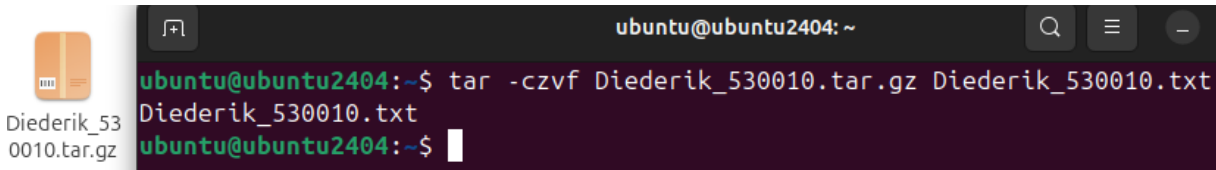
- The /etc directory is typically used for system configuration files.
- It contains settings for the system, services, and applications, such as network configuration, user account info, and startup scripts.

For example:

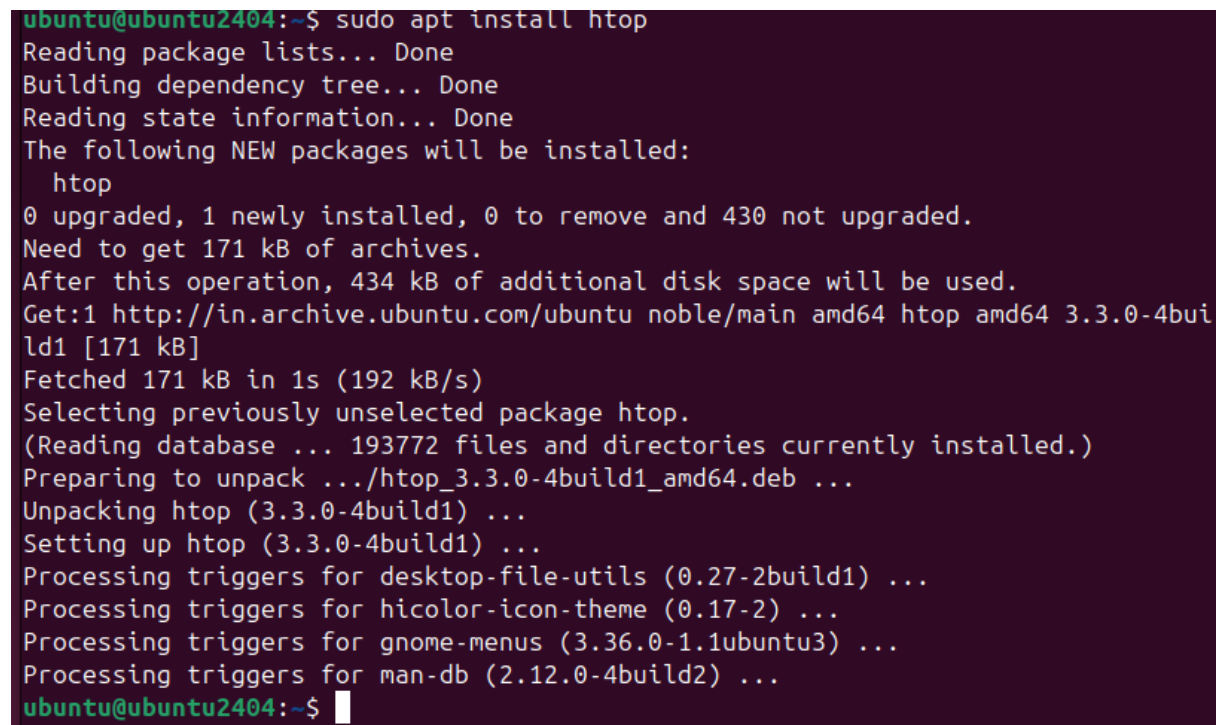
- /etc/passwd → user account info
- /etc/fstab → filesystem mount points
- /etc/ssh/sshd\_config → SSH server configuration

```
tar -cvf archive_name.tar file.txt
```

```
tar -xvf archive_name.tar
```

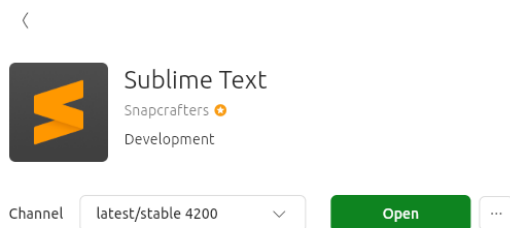


```
ubuntu@ubuntu2404: ~  
ubuntu@ubuntu2404:~$ tar -czvf Diederik_530010.tar.gz Diederik_530010.txt  
Diederik_530010.txt  
ubuntu@ubuntu2404:~$
```



```
ubuntu@ubuntu2404:~$ sudo apt install htop  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following NEW packages will be installed:  
  htop  
0 upgraded, 1 newly installed, 0 to remove and 430 not upgraded.  
Need to get 171 kB of archives.  
After this operation, 434 kB of additional disk space will be used.  
Get:1 http://in.archive.ubuntu.com/ubuntu noble/main amd64 htop amd64 3.3.0-4build1 [171 kB]  
Fetched 171 kB in 1s (192 kB/s)  
Selecting previously unselected package htop.  
(Reading database ... 193772 files and directories currently installed.)  
Preparing to unpack .../htop_3.3.0-4build1_amd64.deb ...  
Unpacking htop (3.3.0-4build1) ...  
Setting up htop (3.3.0-4build1) ...  
Processing triggers for desktop-file-utils (0.27-2build1) ...  
Processing triggers for hicolor-icon-theme (0.17-2) ...  
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
ubuntu@ubuntu2404:~$
```

htop is an interactive system monitor for Linux. It provides a real-time, color-coded view of system processes and resource usage.



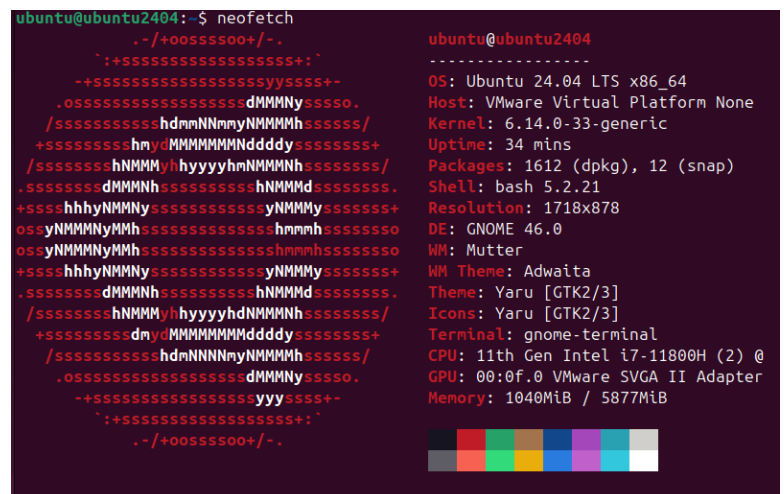
<



**Sublime Text**  
Snapcrafters   
Development

Channel latest/stable 4200 





```
ubuntu@ubuntu2404:~$ neofetch  
      .-/+oosssso+/-.  
      `:+ssssssssssss++`  
      -+ssssssssssssssyyssss+-  
      .oossssssssssssssdMMMMyssso.  
      /ssssssssssshdnnNNnnnyNNMMMyssss+/   
      +ssssssssshmydMMMMMMMNddddyssssss++  
      /ssssssshNMMMyhhyyyhNMMMNhssssss+/   
      .sssssssdMMMNhssssssshNMMMdssssss+.  
      +ssshhhyNNMMNysssssssssyNNMMMyssssss++  
      ossyNNMMNyMMhssssssssshmmhssssssso  
      ossyNNMMNyMMhssssssssshmmhssssssso  
      +ssshhhyNNMMNysssssssssyNNMMMyssssss++  
      .sssssssdMMMNhssssssshNMMMdssssss+.  
      /ssssssshNMMMyhhyyyhNMMMNhssssss+/   
      +sssssssdmydMMMMMMMNddddyssssss++  
      /ssssssshdnnNNnnnyNNMMMyssss+/   
      .oossssssssssssssdMMMMyssso.  
      -+ssssssssssssssyyssss+-  
      `:+ssssssssssss++`  
      .-/+oosssso+/-.  
      _____  
      OS: Ubuntu 24.04 LTS x86_64  
      Host: VMware Virtual Platform None  
      Kernel: 6.14.0-33-generic  
      Uptime: 34 mins  
      Packages: 1612 (dpkg), 12 (snap)  
      Shell: bash 5.2.21  
      Resolution: 1718x878  
      DE: GNOME 46.0  
      WM: Mutter  
      WM Theme: Adwaita  
      Theme: Yaru [GTK2/3]  
      Icons: Yaru [GTK2/3]  
      Terminal: gnome-terminal  
      CPU: 11th Gen Intel i7-11800H (2) @  
      GPU: 00:0f.0 VMware SVGA II Adapter  
      Memory: 1040MiB / 5877MiB
```

neofetch is a system information tool that displays key details about your system in a visually appealing way, often alongside your Linux distribution logo.

## Assignment 5.5: Users and permissions on Linux

Relevant screenshots + motivation

```
ubuntu@ubuntu2404:~$ chmod +x ~/hello/hello.sh
ubuntu@ubuntu2404:~$ cd ~/hello
ubuntu@ubuntu2404:~/hello$ ./hello.sh
Hello Diederik Wiebing 530010!
ubuntu@ubuntu2404:~/hello$ chmod 744 ~/hello/hello.sh
ubuntu@ubuntu2404:~/hello$
```

## Assignment 5.6: View the contents of files

Relevant screenshots + motivation

```
ubuntu@ubuntu2404:~$ wc SherlockHolmes.txt
12306 107562 607504 SherlockHolmes.txt
ubuntu@ubuntu2404:~$
```

```
ubuntu@ubuntu2404:~$ grep -n "kingdom" SherlockHolmes.txt
490:"I tell you that I would give one of the provinces of my kingdom to
1124:And that was how a great scandal threatened to affect the kingdom of
```

```
ubuntu@ubuntu2404:~$ head -n 500 SherlockHolmes.txt | tail -n 20
"Pray do so. I shall be all anxiety."
"Then, as to money?"
"You have _carte blanche_."
"Absolutely?"
"I tell you that I would give one of the provinces of my kingdom to
have that photograph."
"And for present expenses?"
The King took a heavy chamois leather bag from under his cloak and laid
it on the table.
"There are three hundred pounds in gold and seven hundred in notes," he
said.
ubuntu@ubuntu2404:~$
```

```
ubuntu@ubuntu2404:~$ head -n 1134 SherlockHolmes.txt | tail -n 20
The King stared at him in amazement.
"Irene's photograph!" he cried. "Certainly, if you wish it."
"I thank your Majesty. Then there is no more to be done in the matter.
I have the honour to wish you a very good morning." He bowed, and,
turning away without observing the hand which the King had stretched
out to him, he set off in my company for his chambers.
And that was how a great scandal threatened to affect the kingdom of
Bohemia, and how the best plans of Mr. Sherlock Holmes were beaten by a
woman's wit. He used to make merry over the cleverness of women, but I
have not heard him do it of late. And when he speaks of Irene Adler, or
when he refers to her photograph, it is always under the honourable
title of _the_ woman.
II. THE RED-HEADED LEAGUE
ubuntu@ubuntu2404:~$
```

## Assignment 5.7: Digital forensics

Relevant screenshots + motivation

Phone brand/type: motorola/moto g(6) play

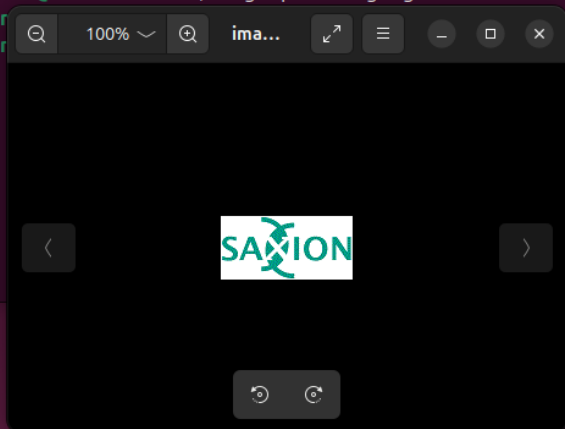
GPS-coördinaten: 53 deg 11' 39.68" N , 6 deg 32' 12.90" E

Stad: Niet te bepalen

```
ubuntu@ubuntu2404:~/Downloads$ file oldcar.  
oldcar.: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment  
length 16, Exif Standard: [TIFF image data, big-endian, direntries=10, manufact  
urer=motorola, model=moto g(6) play, xresolution=160, yresolution=168, resolutio  
nunit=2, software=aljeter-user 9 PPPS29.55-35-18-7 6a0d0 release-keys, datetime=  
2020:11:07 15:08:57, GPS-Data], baseline, precision 8, 4160x3120, components 3  
ubuntu@ubuntu2404:~/Downloads$
```

Yes, ubuntu still considers it as a jpg file

```
ubuntu@ubuntu2404:~$ nano email.txt  
ubuntu@ubuntu2404:~$ base --decode email.txt > image.gif  
Command 'base' not found, did you mean:  
  command 'bake' from snap bake (0.1.3)  
  command 'ase' from deb ase (3.22.1-3)  
  command 'bash' from deb bash (5.2.21-2ubuntu1)  
  command 'basex' from deb basex (10.5-1)  
  command 'basez' from deb basez (1.6.2-3)  
See 'snap info <snapname>' for additional versions.  
ubuntu@ubuntu2404:~$ base64 --decode email.txt > image.gif  
ubuntu@ubuntu2404:~$ file image.gif  
image.gif: GIF image data, version 89a, 108 x 52  
ubuntu@ubuntu2404:~$ xdg-open image.gif  
ubun  
ubun
```



## Assignment 5.8: Steganography

Relevant screenshots + motivation

```
ubuntu@ubuntu2404:~/Downloads$ steghide extract -sf apple2.jpg  
Enter passphrase:  
wrote extracted data to "message.txt".  
ubuntu@ubuntu2404:~/Downloads$ ls  
apple2.jpg  code  code.zip  message.txt  oldcar.  
ubuntu@ubuntu2404:~/Downloads$ cat message.txt  
Hello class.  
You have almost completed Week 5.
```

### Assignment 5.9: Capture disk images

Make relevant screenshots + motivation:

- Proof that the Debian 13 server stored a back-up image of the Ubuntu 24.04 Desktop VM.
- Proof that you can restore the back-up image into an empty VM.

Ready? Save this file and export it as a pdf file with the name: [week5.pdf](#)